

1 Identifying a System from Input/Output Data

(a) Put the input and output samples into the matrices

$$X = \begin{bmatrix} x^{(1)} & \dots & x^{(N)} \end{bmatrix}, \quad Y = \begin{bmatrix} y^{(1)} & \dots & y^{(N)} \end{bmatrix}$$

Then

$$J = \sum_{k=1}^N \|Ax^{(k)} - y^{(k)}\|^2 = \|AX - Y\|_F^2$$

The least-squares normal equation is

$$AXX^T = YX^T$$

If X has full row rank, then XX^T is invertible, and the estimate is

$$\hat{A} = YX^T(XX^T)^{-1}$$

This is the same as solving one least-squares problem for each row of A .

(b) For this dataset $m = 10$, $n = 3$, $N = 100$ and $\text{rank}(X) = 3$ so X has full row rank and XX^T is invertible.

```
Ahat =
10×3 Matrix{Float64}:
 2.0299  5.0208  5.0104
 0.0114  6.9999  1.0106
 7.0424 -0.0025  6.9448
 6.9977  3.9759  4.0024
 9.013   1.0449  6.998
 4.0119  3.9649  9.0267
 4.9871  6.9723  8.0336
 7.9425  6.0875  3.0174
 0.0094  8.9722 -0.0385
 1.0612  8.0208  7.0285
RMS residual = 6.453362
Frobenius residual = 64.533618
```

1.1 Code

```
1 using LinearAlgebra
2
3 include(joinpath(@__DIR__, "readclassjson.jl"))
4
5 data = readclassjson(joinpath(@__DIR__, "sysid_data.json"))
6 X = data["X"]
7 Y = data["Y"]
8
9 Ahat = Y * X' * inv(X * X')
10 residual = Ahat * X - Y
```

2 Simultaneous Left Inverse of Two Matrices

We need $H \in \mathbb{R}^{2 \times 5}$ such that

$$HG = I, \quad H\tilde{G} = I$$

An explicit matrix that works is

$$H = \begin{bmatrix} -18/25 & 71/25 & 16/25 & -2/5 & 0 \\ 8/25 & -1/25 & 4/25 & -3/5 & 0 \end{bmatrix}$$

Direct multiplication gives

$$HG = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad H\tilde{G} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Therefore this H recovers x from either measurement vector.

```
H =
2x5 Matrix{Rational{Int64}}:
 -18//25  71//25  16//25  -2//5  0
  8//25  -1//25  4//25  -3//5  0
H*G =
2x2 Matrix{Rational{Int64}}:
 1  0
 0  1
H*Gtilde =
2x2 Matrix{Rational{Int64}}:
 1  0
 0  1
```

2.1 Code

```
1 G = [2//1 3//1;
2       1//1 0//1;
3       0//1 4//1;
4       1//1 1//1;
5       -1//1 2//1]
6
7 Gtilde = [-3//1 -1//1;
8            -1//1 0//1;
9            2//1 -3//1;
10           -1//1 -3//1;
11            1//1 2//1]
12
13 H = [-18//25 71//25 16//25 -2//5 0//1;
14       8//25 -1//25 4//25 -3//5 0//1]
15
16 H * G
17 H * Gtilde
```

3 Optimal Flow on a Data Collection Network

(a) With the incidence convention in the problem, $(Af)_i$ is the flow entering node i minus the flow leaving node i . Conservation gives

$$(Af)_i + s_i = 0$$

for each regular node, so

$$Af = -s$$

The optimization problem is therefore

$$\min_f \frac{1}{n} \|f\|_2^2 \quad \text{subject to} \quad Af = -s$$

The factor $1/n$ does not change the optimizer. Since the graph is connected to the destination node, the reduced incidence matrix A has linearly independent rows. Thus AA^T is invertible, and the minimum-norm feasible flow is

$$f^* = -A^T(AA^T)^{-1}s$$

(b) Using the arc directions in the figure,

$$A = \begin{bmatrix} -1 & 1 & -1 & -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & -1 \end{bmatrix}$$

```
A*fopt =
4-element Vector{Float64}:
 -1.0
 -4.0
 -10.0
 -10.0
fopt =
7-element Vector{Float64}:
 11.9524
  4.619
 -1.0952
 -5.2381
 -0.619
  4.1429
 13.0476
mean square optimal flow = 54.374150
mean square simple flow = 77.285714
```

So the optimal flow has a smaller mean-square traffic level.

3.1 Code

```
1 using LinearAlgebra
2
3 A = [-1.0  1.0 -1.0 -1.0  0.0  0.0  0.0;
4      0.0 -1.0  0.0  0.0 -1.0  0.0  0.0;
```

```
5      0.0  0.0  0.0  1.0  1.0 -1.0  0.0;  
6      0.0  0.0  1.0  0.0  0.0  1.0 -1.0]  
7  
8  s = [1.0, 4.0, 10.0, 10.0]  
9  fsimple = [5.0, 4.0, 0.0, 0.0, 0.0, 10.0, 20.0]  
10  
11  fopt = -A' * inv(A * A') * s  
12  mean_square_opt = sum(fopt.^2) / length(fopt)  
13  mean_square_simple = sum(fsimple.^2) / length(fsimple)
```

4 Optimal Control of Unit Mass

For each interval $i - 1 < t \leq i$, the force is constant:

$$f(t) = x_i$$

With unit mass, $p''(t) = f(t)$. Over one unit interval,

$$\dot{p}_i = \dot{p}_{i-1} + x_i, \quad p_i = p_{i-1} + \dot{p}_{i-1} + \frac{1}{2}x_i$$

Thus, for an integer time k ,

$$\dot{p}(k) = \dot{p}(0) + \sum_{j=1}^k x_j$$

and

$$p(k) = p(0) + k\dot{p}(0) + \sum_{j=1}^k \left(k - j + \frac{1}{2}\right) x_j$$

(a) Here $p(0) = \dot{p}(0) = 0$. The objective is

$$\int_0^{10} f(t)^2 dt = \sum_{i=1}^{10} x_i^2 = \|x\|_2^2$$

The constraints $p(10) = 1$, $\dot{p}(10) = 0$, and $p(5) = 0$ can be written as

$$Cx = d$$

where the rows of C are the coefficient rows for $p(10)$, $\dot{p}(10)$, and $p(5)$, which are linearly independent, so C has full rank and CC^T is invertible. Therefore the minimum-energy force vector is the minimum-norm solution

$$x^* = C^T(CC^T)^{-1}d$$

```
1part (a) x =
10-element Vector{Float64}:
 -0.045455
 -0.007576
  0.030303
  0.068182
  0.106061
  0.093939
  0.031818
 -0.030303
 -0.092424
 -0.154545
[p(10), pdot(10), p(5)] =
3-element Vector{Float64}:
 1.0
 0.0
 0.0
part (a) energy = 0.062121
```

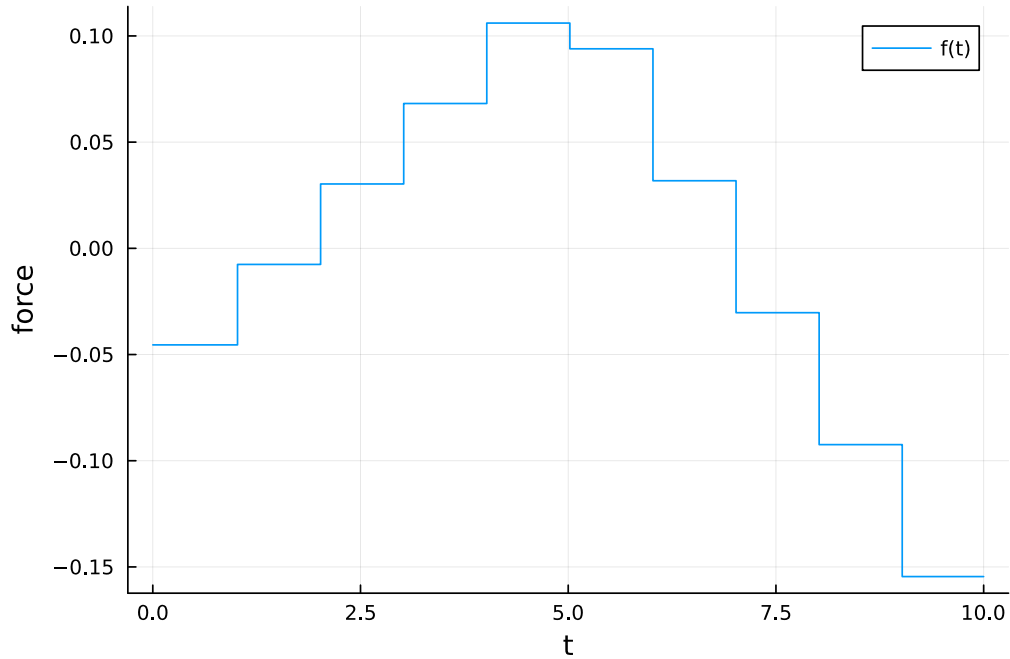


Figure 1: Optimal piecewise-constant force for part (a)

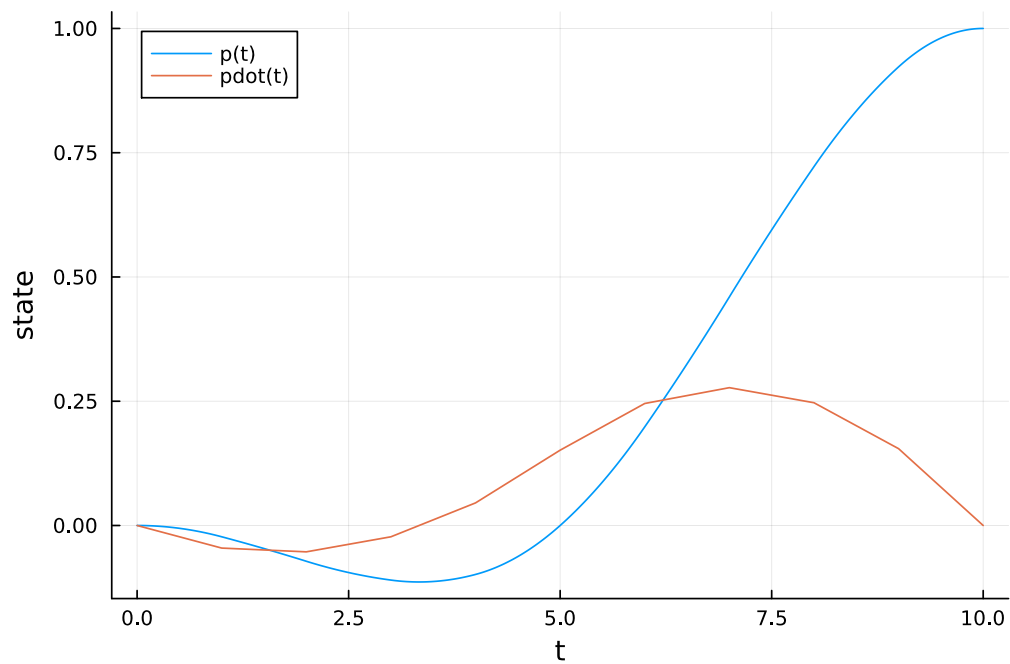


Figure 2: Position and velocity produced by the optimal force

The force first keeps the mass near the origin so that the midpoint constraint $p(5) = 0$ is met, and then it pushes the mass toward $p(10) = 1$. Near the end it applies negative force to bring the velocity back to zero.

(b) Now $p(0) = 0$ and $\dot{p}(0) = 1$. Let

$$z = \begin{bmatrix} p(10) \\ \dot{p}(10) \end{bmatrix} = Bx + b$$

where $b = [10 \ 1]^T$ is the terminal state with zero force. Then

$$J_1 = \|Bx + b\|_2^2, \quad J_2 = \|x\|_2^2$$

For each $\mu > 0$, I minimize

$$J_1 + \mu J_2 = \|Bx + b\|_2^2 + \mu \|x\|_2^2$$

which gives

$$x_\mu = -(B^T B + \mu I)^{-1} B^T b$$

Endpoints:

```
part (b) J1=0 endpoint: J1 = 1.97215e-31, J2 = 0.403030
part (b) J2=0 endpoint: J1 = 101.000000, J2 = 0
```

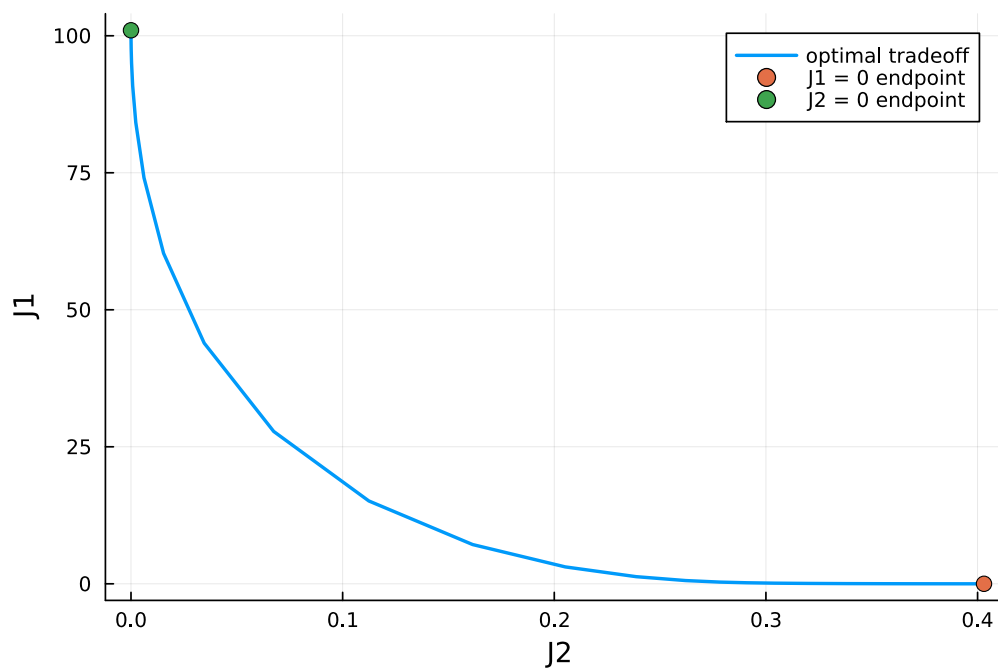


Figure 3: Optimal trade-off between terminal error J_1 and force energy J_2

4.1 Code

```
1 using LinearAlgebra
2 using Plots
3
4 p10 = [10 - j + 0.5 for j in 1:10]
5 v10 = ones(10)
6 p5 = [j <= 5 ? 5 - j + 0.5 : 0.0 for j in 1:10]
```

```
7
8 C = [p10'; v10'; p5']
9 d = [1.0, 0.0, 0.0]
10 x_a = C' * inv(C * C') * d
11
12 B = [p10'; v10']
13 b = [10.0, 1.0]
14 mu_values = 10.0 .^ range(-6, 6, length=50)
15 x_mu = [-(B' * B + mu * I) \ (B' * b) for mu in mu_values]
16 J1 = [norm(B * x + b)^2 for x in x_mu]
17 J2 = [sum(x.^2) for x in x_mu]
```


5 Modifying Measurements to Satisfy Known Conservation Laws

Let

$$A = \begin{bmatrix} a_1 & \cdots & a_m \end{bmatrix} \in \mathbb{R}^{n \times m}, \quad b = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

The conservation laws are

$$A^T x = b$$

We want the smallest correction c such that

$$A^T(y + c) = b$$

Equivalently,

$$A^T c = b - A^T y$$

The minimum-norm solution of this linear equation is

$$c = A(A^T A)^{-1}(b - A^T y)$$

The inverse exists because $a_1, \dots, a_m$ are independent, so A has linearly independent columns. Therefore $A^T A$ is invertible.

The adjusted measurement is

$$y_{\text{adj}} = y + A(A^T A)^{-1}(b - A^T y)$$

Now verify the conservation laws:

$$A^T y_{\text{adj}} = A^T y + A^T A(A^T A)^{-1}(b - A^T y)$$

$$A^T y_{\text{adj}} = A^T y + b - A^T y = b$$

So the adjusted vector satisfies all known conservation laws exactly.

6 Invertibility of the KKT Matrix

Let

$$K = \begin{bmatrix} A^T A & C^T \\ C & 0 \end{bmatrix}, \quad S = \begin{bmatrix} A \\ C \end{bmatrix}$$

(a) For any u ,

$$Su = 0$$

if and only if

$$Au = 0, \quad Cu = 0$$

Therefore

$$\text{null}(S) = \text{null}(A) \cap \text{null}(C)$$

The matrix S has linearly independent columns if and only if $\text{null}(S) = \{0\}$. Hence S has linearly independent columns if and only if

$$\text{null}(A) \cap \text{null}(C) = \{0\}$$

(b) If C does not have linearly independent rows, then there is a nonzero λ such that

$$C^T \lambda = 0$$

Then

$$K \begin{bmatrix} 0 \\ \lambda \end{bmatrix} = \begin{bmatrix} C^T \lambda \\ 0 \end{bmatrix} = 0$$

The vector $[0 \ \lambda]^T$ is nonzero, so K has a nonzero null vector and is singular.

(c) Suppose there exists a nonzero vector $u \in \text{null}(A) \cap \text{null}(C)$. Then

$$Au = 0, \quad Cu = 0$$

and

$$K \begin{bmatrix} u \\ 0 \end{bmatrix} = \begin{bmatrix} A^T Au \\ Cu \end{bmatrix} = 0$$

The vector $[u \ 0]^T$ is nonzero, so K is singular.

(d) Assume K is singular. Then there exists a nonzero pair (u, λ) such that

$$\begin{bmatrix} A^T A & C^T \\ C & 0 \end{bmatrix} \begin{bmatrix} u \\ \lambda \end{bmatrix} = 0$$

This means

$$A^T Au + C^T \lambda = 0, \quad Cu = 0$$

Multiply the first equation on the left by u^T :

$$u^T A^T Au + u^T C^T \lambda = 0$$

Since $Cu = 0$,

$$u^T C^T \lambda = (Cu)^T \lambda = 0$$

so

$$\|Au\|_2^2 = 0$$

and therefore

$$Au = 0$$

Thus $u \in \text{null}(A) \cap \text{null}(C)$. If $u \neq 0$, then there is a nonzero vector in this intersection.

If $u = 0$, then the pair (u, λ) is nonzero only if $\lambda \neq 0$. The first KKT equation becomes

$$C^T \lambda = 0$$

with $\lambda \neq 0$, so the rows of C are linearly dependent.

Therefore, if K is singular, then either C does not have linearly independent rows, or there exists a nonzero vector

$$u \in \text{null}(A) \cap \text{null}(C)$$